

Computational Intelligence

Report Lab3 (CNN)

Author: Jakub Jabłoński

Section 3.1.3

Perform sentiment classification using IMDB dataset

- a) Report results in the table summarizing parameter values and classification results (report accuracy, precision, recall, F1).

For this configs I used 2 values of specific data:

- Larger (example: embed_dim: 128)
- Smaller (example: embed_dim: 64)

(Just to be specific) In the next tasks, I will use the term “Larger” for greater values and “Smaller for the smaller values.

```
configs_dict = [  
    {"Conf_name": "C1", "seq_len": 128, "vocab_size": 10000,  
    "embed_dim": 64, "heads": 2, "ff_dim": 128, "epochs": 5, "lr": 1e-3},  
    {"Conf_name": "C2", "seq_len": 128, "vocab_size": 10000,  
    "embed_dim": 128, "heads": 4, "ff_dim": 256, "epochs": 5, "lr": 1e-3},  
    {"Conf_name": "C3", "seq_len": 256, "vocab_size": 10000,  
    "embed_dim": 64, "heads": 4, "ff_dim": 256, "epochs": 10, "lr": 1e-3},  
    {"Conf_name": "C4", "seq_len": 256, "vocab_size": 10000,  
    "embed_dim": 128, "heads": 4, "ff_dim": 256, "epochs": 5, "lr": 5e-4},  
    {"Conf_name": "C5", "seq_len": 128, "vocab_size": 10000,  
    "embed_dim": 128, "heads": 2, "ff_dim": 128, "epochs": 10, "lr": 1e-3},  
    {"Conf_name": "C6", "seq_len": 256, "vocab_size": 10000,  
    "embed_dim": 64, "heads": 4, "ff_dim": 128, "epochs": 10, "lr": 5e-4},  
    {"Conf_name": "C7", "seq_len": 256, "vocab_size": 10000,  
    "embed_dim": 128, "heads": 4, "ff_dim": 256, "epochs": 10, "lr": 5e-4},  
    {"Conf_name": "C8", "seq_len": 128, "vocab_size": 10000,  
    "embed_dim": 64, "heads": 2, "ff_dim": 128, "epochs": 5, "lr": 5e-4}  
]
```

	accuracy	precision	recall	f1
C1	0.83224	0.820398	0.85072	0.835284
C2	0.85736	0.842982	0.87832	0.860288
C3	0.85776	0.837357	0.88800	0.861935
C4	0.81788	0.774698	0.89648	0.831151
C5	0.85532	0.830986	0.89208	0.860450
C6	0.84232	0.827591	0.86480	0.845787
C7	0.85028	0.836834	0.87024	0.853210
C8	0.78704	0.762204	0.83440	0.796670

- **Accuracy:**

Configs C2 and C3 have the best accuracy, on the other hand, the worst accuracy is attributed to C8. The C2 and C3 configurations have values of **ff_dim**= 256 and **heads** = 4, while the C8 configuration has **ff_dim**= 256 and **heads** = 4. This shows that these parameters have the greatest impact on model accuracy.

- **Precision:**

The situation is almost identical, with the difference that C2 has better precision than C3 this time. The C2 configuration achieves the best result, while the C4 and C8 configurations have the worst results. That shows that the larger the embedding dimension, the greater the precision.

- **Recall:**

The C4 and C5 configurations had the highest recall, while the C8 and C1 configurations had the lowest. The C4 configuration had a lower learning rate, and the C5 configuration had more epochs, but both configurations had the same embedded dimensions.

- **F1:**

Configuration C3 wins in parameter F1, followed closely by configurations C2 and C5. Configuration C3 has a larger number of

heads and higher feed-forward dimension. It also has a greater number of epochs and a larger embedding dimension. The worst f1 score belongs to C8, which apparently has the worst parameters and is under-trained.

b) Display a few reviews from the dataset and print classification result

(!!) Results in the notebook (!!)

```
Running C1...
Test text: Simply delightful claymation feature from Nick Park and company spoofing such film-greats as "King Kong" and "The Wolf Man"
Prediction: Negative True value: Negative
Score: 0.231

Running C2...
Test text: I've never given a movie a ten out of ten before but this is the closest I have ever come (I gave it a 9). There are very f
Prediction: Positive True value: Positive
Score: 0.9216

Running C3...
Test text: I enjoyed this series, but felt that the whole thing was let down by the sound recording/mixing.<br /><br />For whatever re
Prediction: Positive True value: Positive
Score: 0.9958

Running C4...
Test text: First of all, i must say i really enjoy watching this movie. The way we follow the different people at the same time works
Prediction: Positive True value: Positive
Score: 0.6132

Running C5...
Test text: I tried to give this show a chance, but it really doesn't sit well with me. Although the performances are good, the writing
Prediction: Negative True value: Negative
Score: 0.0076

Running C6...
Test text: If you want to see a film with no guns, blood, sex, shouting angry people, hero, bad guys & girls or even clumsy love words
Prediction: Negative True value: Negative
Score: 0.0223

Running C7...
Test text: My entire family enjoyed this film, including 2 small children. Great values without sex, violence, drugs, nudity, or profa
Prediction: Negative True value: Negative
Score: 0.0149

Running C8...
Test text: In New York, a group of freshmen join the High School for the Performing Arts after being well succeeded in their audition.
Prediction: Negative True value: Negative
Score: 0.1678
```

Section 3.2.4

a) Test at least 4 configurations using TF or Pytorch

To prevent the test from taking forever, I reduced the number of epochs to 5.

```
configs_vit = [  
  
{"name": "C1", "depth": 2, "emb_size": 64, "patch_size": 4, "use_cls_token": True},  
  
{"name": "C2", "depth": 4, "emb_size": 128, "patch_size": 4, "use_cls_token": True},  
  
{"name": "C3", "depth": 6, "emb_size": 128, "patch_size": 2, "use_cls_token": False},  
  
{"name": "C4", "depth": 8, "emb_size": 256, "patch_size": 8, "use_cls_token": True},  
]
```

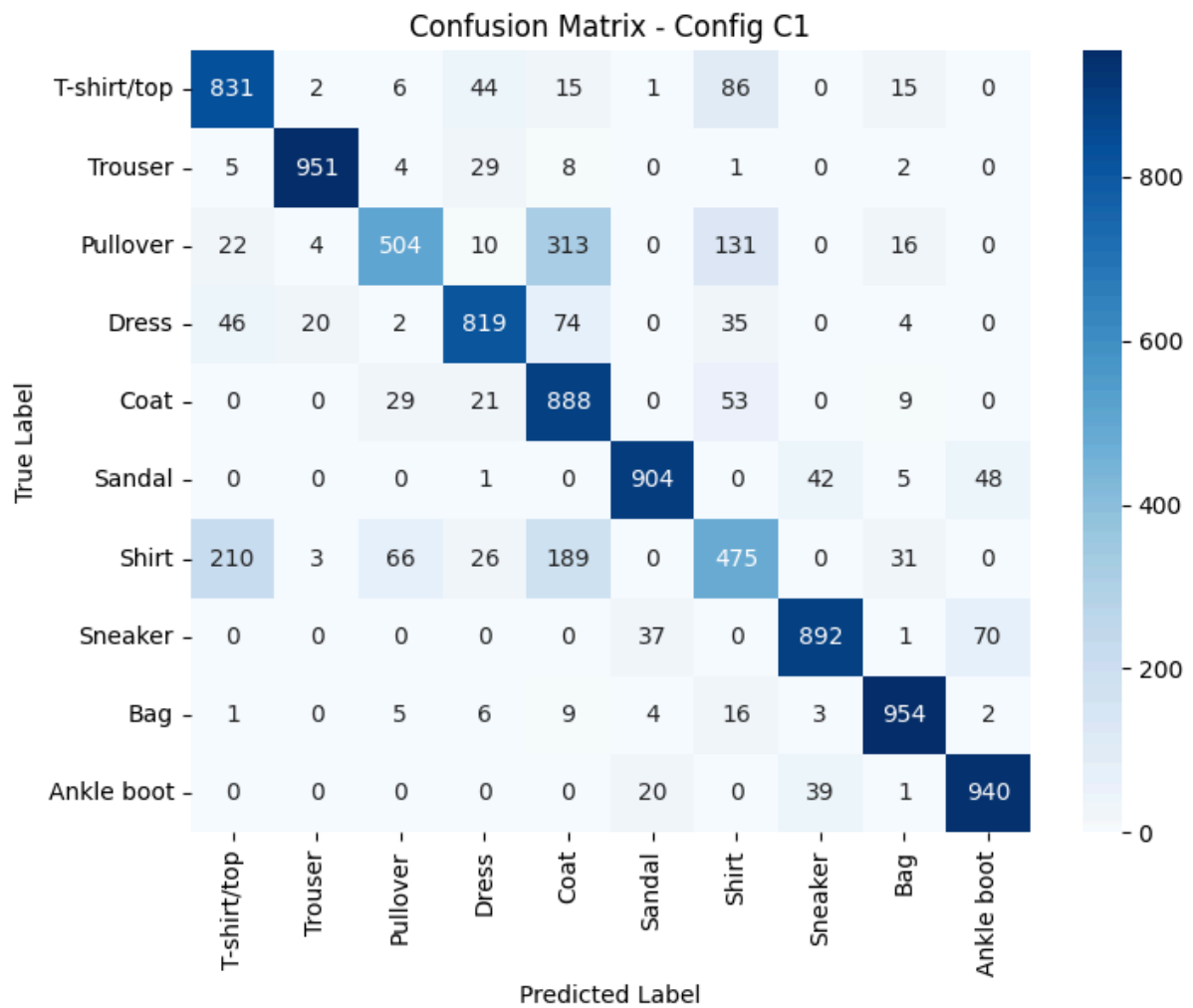
Results of trainings:

	Config	Accuracy	Precision_T-shirt/top	Recall_T-shirt/top	F1_T-shirt/top	Precision_Trouser	Recall_Trouser	F1_Trouser
0	C1	0.8356	0.782522	0.788	0.785252	0.989507	0.943	0.965694
1	C2	0.8654	0.847420	0.772	0.807954	0.972056	0.974	0.973027
2	C3	0.8502	0.868932	0.716	0.785088	0.989669	0.958	0.973577
3	C4	0.8547	0.794747	0.817	0.805720	0.983690	0.965	0.974255

(More results are in my notebook)

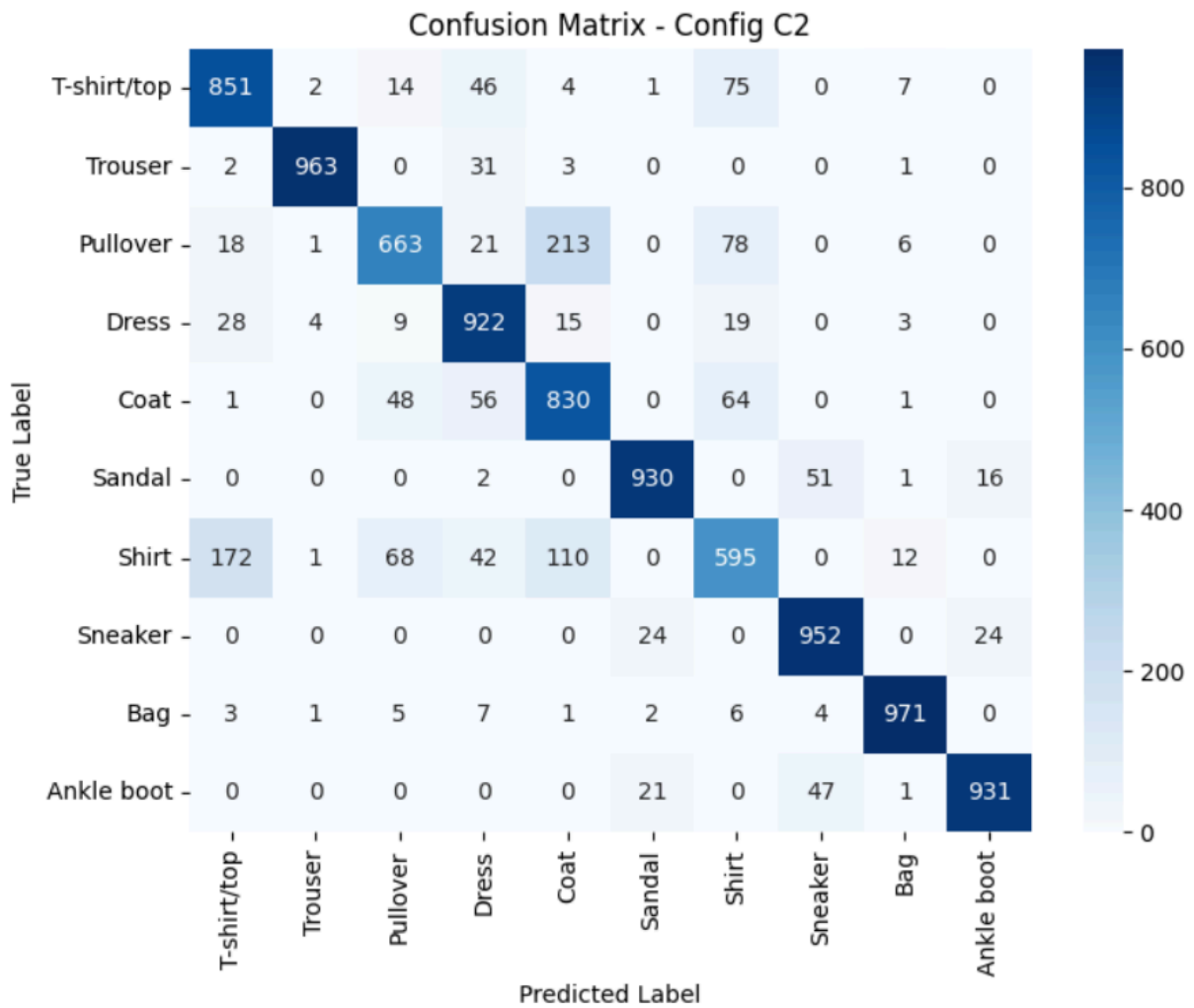
b) Include confusion matrix (use correct labels) and discuss which images were misclassified (e.g. aircrafts taken as frogs?)

- C1



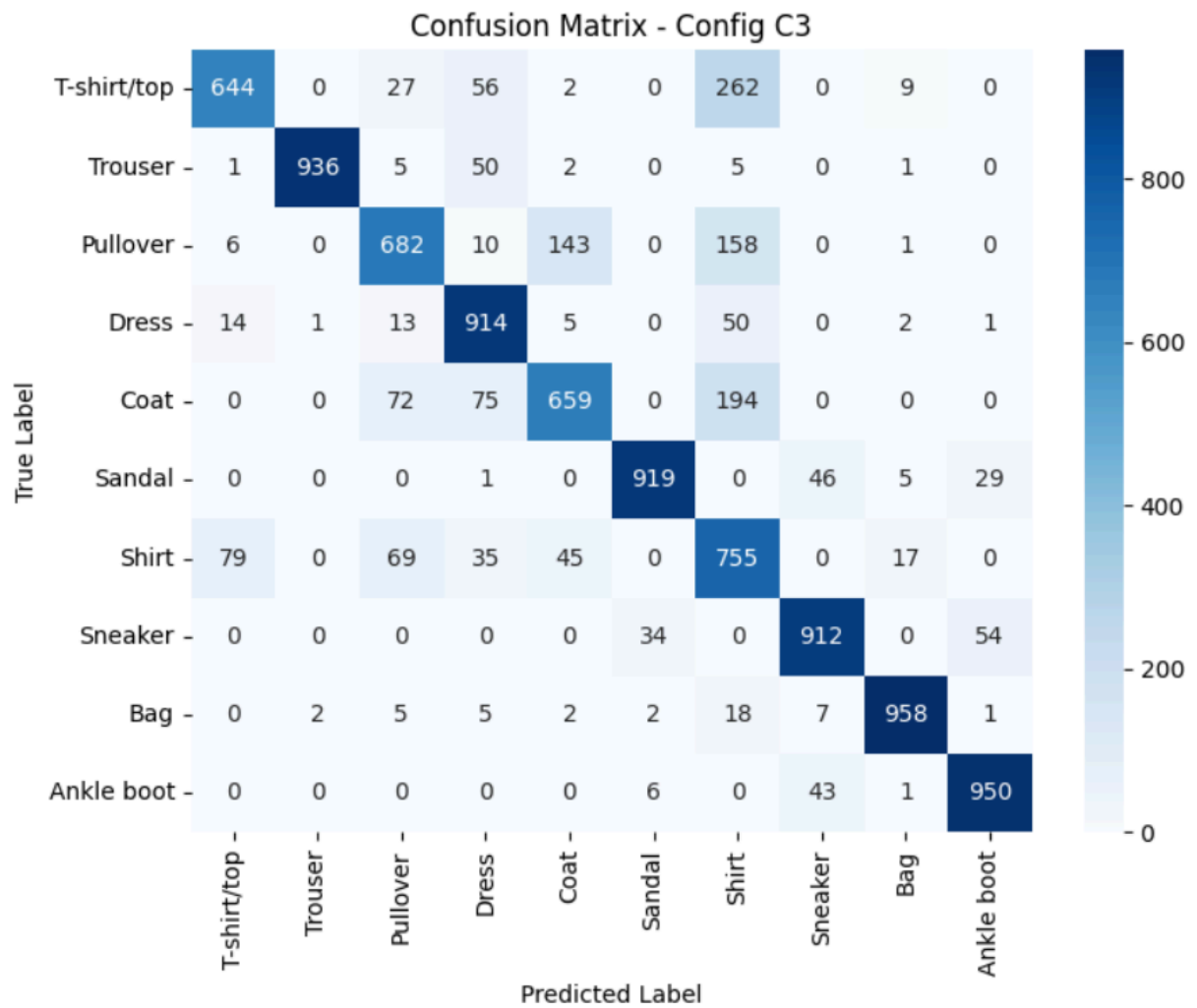
It seems that the model often confused a coat with a pullover, and a T-shirt with a shirt.

- C2



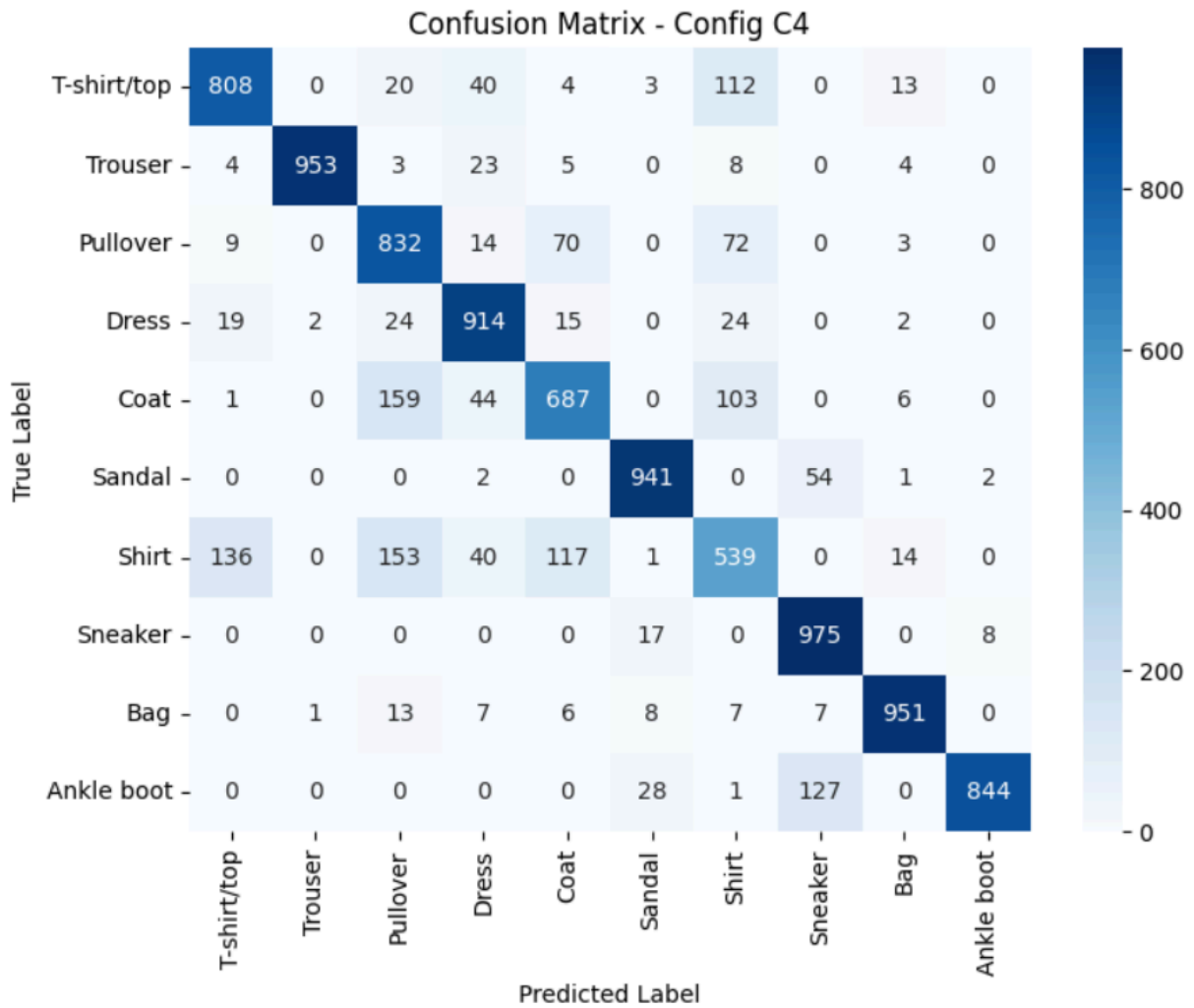
The situation here is similar to that in C1, only this time the model performed better in terms of classification.

- C3



Here, the model often stated that a shirt is a coat, a pullover, and a T-shirt and that the coat is a pullover.

- C4



The situation was the same as in configuration C3, but this time the model confused coats, pullovers, and T-shirts with shirts much more often.

Section 3.3.3

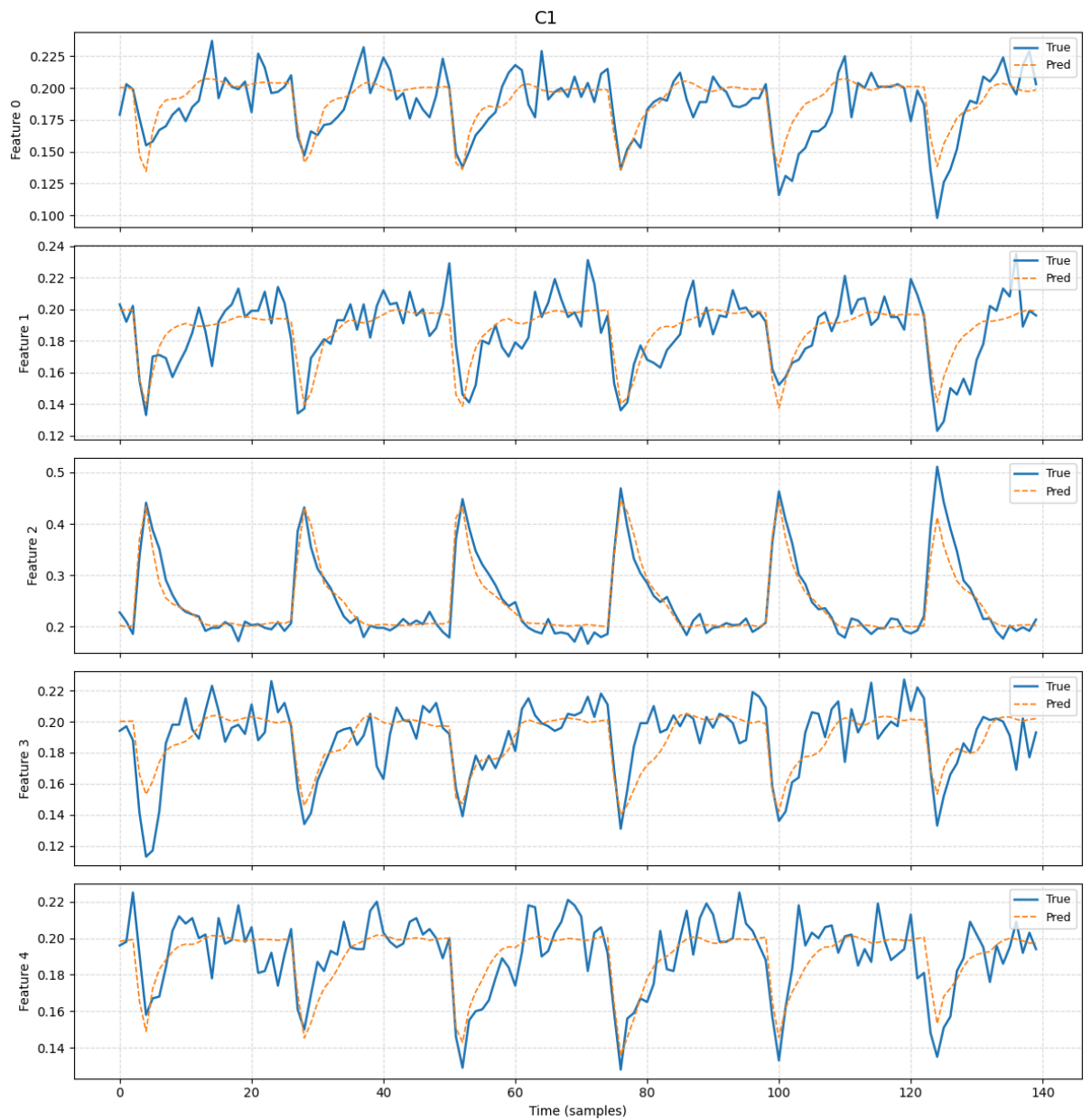
- a) Give summary results in a table. Include parameters and regression scores. Add selected plots (e.g. for the best and worst results).

Model was trained with 20 epochs, to save some time.

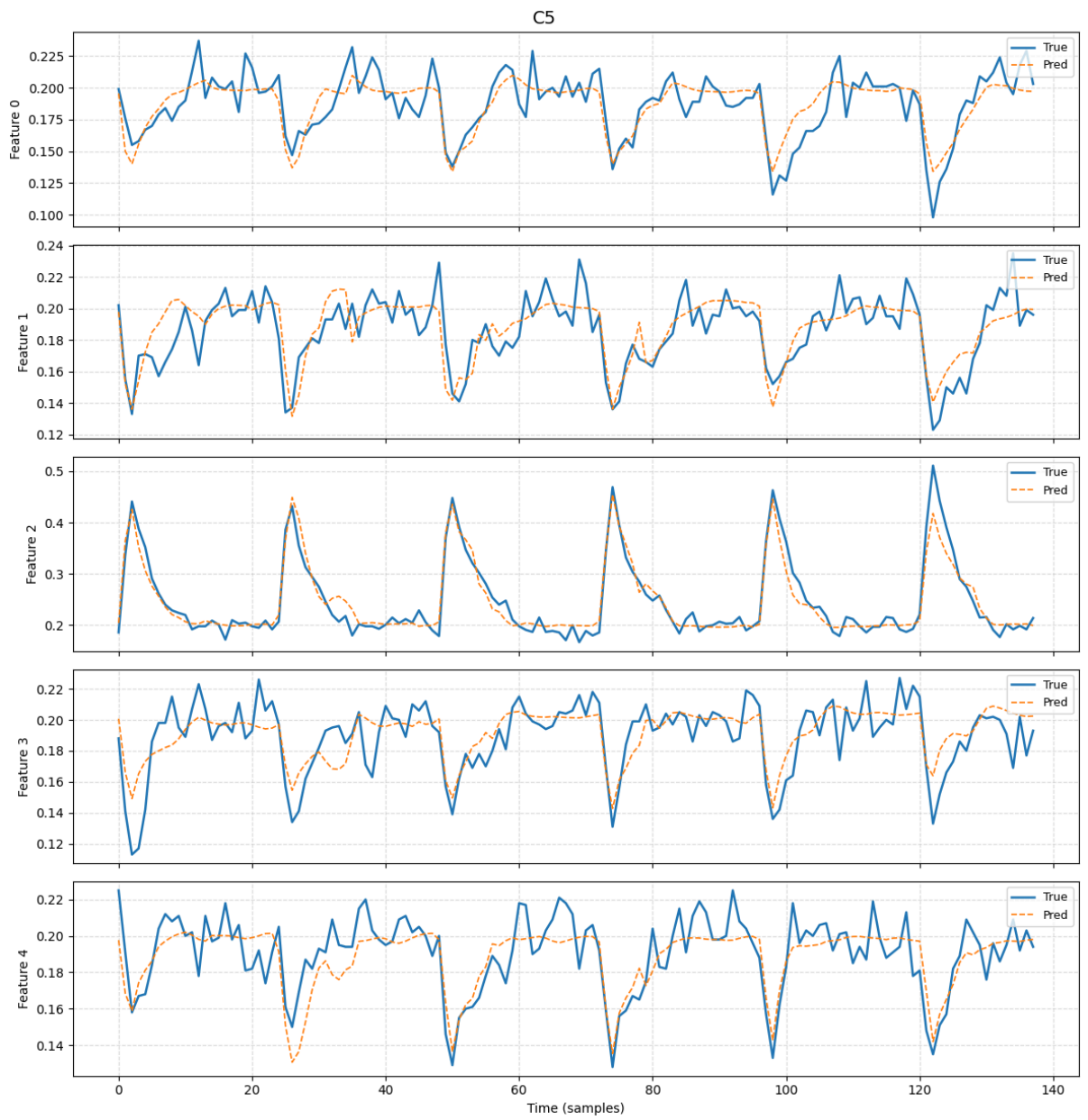
	window	d_model	heads	layers	ff_dim	MSE	RMSE	MAE	R2
0	32	128	4	2	256	0.000253	0.015895	0.012007	0.684273
1	50	64	4	4	128	0.000234	0.015291	0.011435	0.682938
2	24	32	2	2	64	0.000276	0.016617	0.012340	0.665918
3	40	128	8	6	64	0.000326	0.018060	0.012263	0.642429
4	4	64	8	6	256	0.000456	0.021363	0.013492	0.600838
5	12	128	8	4	256	0.000548	0.023413	0.014170	0.543382

Smaller window sizes can underfit (Config 4), while large model depth, number of heads or number of layers do not guarantee better performance and may hurt generalization. It looks like config 0 has the most balanced attention parameters and window size compared to other configs.

Best results:



Worst results:



Answer the questions

- a) **What is the complexity of attention mechanism with respect to n - the sequence lengths**

$$O(n^2)$$

Every Q (Query) has to be compared with every K (key) in the sequence.

- b) **Is the attention map global or local (pertaining to the sequence).**

In a standard transformer, the attention map is global. Each token can refer to any other token in the sequence. For example, in CNN, kernels only look at a local fragment of the image, while in transformers have a global view on the image (or sequence).

- c) **Which parameters are learned during training the Transformers models for NLP?**

- **Weight matrices** - Used for calculation Q, K, V variables.
- **Embedding matrices and positional embeddings**
- **Final linear layer (weights and biases)**
- **FFN (weights and biases)**

- d) **What is the role of positional encoding?**

Transformer does not have a built-in concept of order, because attention treats a sequence as a set of independent tokens. Positional encoding adds information about the position of a token in a sequence. It's usually an additional value in the token embedding vector (for example, sinusoidal vectors).

- e) **Which output data resulting from Transformer Block can be used for classification or regression**

We can take the special CLS token and use it for classification, or we can take the mean of the vector of value of hidden states (output of transformer block)